

Yangtao Ge

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EDUCATION

- **University of Oxford** Oxford, UK
Master (Msc) in Software and Systems Security, Part-Time Sept. 2024 – Present
- **University College London** London, UK
Master of Engineering (MEng) in Computer Science; Degree Award: First Class Honours Sept. 2018 – Sept. 2022
- **University College London** London, UK
Undergraduate Preparatory Certificate; Avg. Score 86/100 Sept. 2017 – June. 2018

PROFESSIONAL SUMMARY

Senior Backend Software Engineer specializing in distributed data processing systems and high-throughput data pipelines. Expert in building Python-based asynchronous data processing systems using Apache Arrow for Parquet file operations, achieving 70% latency reduction while processing 10M+ records. Experienced with real-time stream processing pipelines with Apache Flink and AWS Kinesis, implementing efficient data tiering between hot and cold storage. Proven track record in optimizing data infrastructure costs by 50% through strategic system redesign and leveraging columnar data formats. Combines deep expertise in data engineering with cloud-native technologies to deliver scalable, cost-effective solutions for enterprise-scale data challenges.

PERSONAL SKILLS

- **Programming Languages:**
 - **Python:** Proficient in building distributed asynchronous systems data pipelines. Experienced with Apache Arrow for efficient Parquet file operations, multiprocessing for parallel data transformations
 - **Golang:** Proficient in building concurrent applications, primarily for microservices using Chi and Gin frameworks.
 - **Java:** Proficient in building scalable server-side applications using Spring Framework and Gradle. Experienced with Apache Flink for stream processing and real-time data ingestion from AWS Kinesis.
 - **Other Languages:** C/C++ (Competent); JavaScript (Proficient); SQL (Proficient)
- **Cloud Platform & Tools:**
 - **AWS:** Experienced with EC2, SQS, S3, and Lambda for cloud-based applications building with **terraform**
 - **Docker:** Strong capabilities in containerization and version control of deployment environments.
 - **Kubernetes & ECS:** Proficient in orchestrating and managing containerized applications in AWS EKS and ECS
 - **Apache Flink & Kinesis:** Experienced in building real-time stream processing pipelines with Apache Flink and AWS Kinesis for high-throughput data ingestion
- **DevOp & Analytic:**
 - **GitLab/GitHub:** Proficient with git for version control and building CI/CD pipelines with GitLab Pipeline.
 - **OpenSearch:** Implemented for search and log analysis solutions using Fluentd in containerized environments.
 - **Datadog & Grafana:** Proficient in building interactive dashboards for real-time data visualization and analytics.

WORK EXPERIENCE

- **Smartsheet** London, UK
Software Engineer II Jan. 2023 - Present
 - **Asynchronous Distributed Data Processing System:**
 - * Architected and led Python-based distributed asynchronous system with multiple workers orchestrated through AWS SQS and DynamoDB distributed lock tables, eliminating API gateway timeout errors and reducing system latency by 70%.
 - * Implemented Apache Arrow for efficient Parquet file read/write operations with concurrent data transformations, processing 10M+ history records for enterprise "whale grids" while maintaining optimal memory usage.

- * Designed Python distributed processing pipeline with SQS message queuing and DynamoDB for coordination, ensuring data integrity through Apache Arrow's columnar processing capabilities.
- * Achieved 50% cost reduction through strategic system redesign and legacy system deprecation, resolving two critical customer escalations affecting enterprise clients.
- **Real-Time Event Ingestion Pipeline:**
 - * Built high-throughput data ingestion pipeline using Apache Flink to process AWS Kinesis streams, handling millions of events per second with sub-second latency.
 - * Implemented dual data paths architecture: writing to S3 as Parquet files for cost-effective cold storage while simultaneously ingesting to DynamoDB for low-latency hot data access.
 - * Designed stream processing solution that improved data availability by 40% while reducing storage costs through intelligent tiering between hot and cold data paths.
- **DevOps Excellence & Infrastructure:**
 - * Designed and implemented company-wide centralized ECS pipeline (adopted as part of PEAK initiative), reducing service onboarding time from weeks to under 24 hours through template standardization and process automation.
 - * Pioneered adoption of Graviton instances and multi-architecture builds, reducing ECS operational costs by 25% while maintaining performance.
 - * Created hardened, secure container images with automated vulnerability scanning, establishing new security standards for the organization.
- **Spotify** London, UK
Backend Engineer Intern *June. 2022 - Sept. 2022*
 - **Java Microservices:** Developed backend microservices using the Apollo framework and gRPC, which enhances music delivery processes and service efficiency for both music providers and Spotify operatives.
 - **Testing Enhancement:** Authored technical documentation outlining key strategies for testing improvements, which was instrumental in increasing code coverage from 75% to 90%, substantially boosting application reliability
 - **Operational Integration:** Engineered a Slack notification workflow that automatically informs Spotify operatives of stakeholder status updates, enhancing operational communication and responsiveness.
- **R3** London, UK
Solution Engineer *June. 2021 - Sept. 2021*
 - **Confidential Computing Development:** Actively contributed to the enhancement of the Conclave example codebase utilizing Kotlin and Java, which has been utilized in commercial sales processes, communicating with non-technical stakeholders to demonstrate practical applications of confidential computing in business solutions
 - **Corda Certification:** Successfully passed Corda Developer Certification Exam, validating expertise in enterprise blockchain technology such as distributed ledger and contracts. [Credly Credential]

PROJECTS

- **Anxiety Trigger Detection System** Open Lab, Newcastle University, UK
Lead Developer *Oct. 2024 - Sept. 2025*
 - **Privacy-Focused AI Architecture:** Designed and implemented a privacy-first social media analysis system using Python and YOLO object detection models to anonymize sensitive information (usernames, avatars, timestamps) server-side before LLM processing. Deployed custom-trained YOLO models within AWS Lambda for real-time PII detection and masking.
 - **Prompt Engineering & Multi-Modal Analysis:** Developed sophisticated prompt engineering framework enabling researchers to test different prompt configurations across multiple AI providers (OpenAI, AWS Bedrock, Claude) with user-specific customization. Implemented multi-modal analysis pipeline processing both image and text content to identify anxiety triggers with language-specific prompts for English and Chinese.
 - **Serverless Python Pipeline:** Built event-driven processing pipeline using Python Lambda functions, EventBridge, and SQS for asynchronous image processing. Integrated YOLO models for object detection, achieving sub-second anonymization while maintaining 99% accuracy in PII identification across light/dark theme variations.
- **Fuzz Testing for DPC++** University College London, UK
Individual Project Researcher *Sept. 2021 - Jun. 2022*
 - **DPC++ Fuzzing Framework:** Wrote a dissertation to propose the first fuzzing framework for DPC++ which includes an instrument tool to track the coverage and a fuzzer to generate test inputs. Establish a DPC++ code collection to benchmark the testing framework effectiveness.
 - **LLVM & Clang:** Implement the testing framework under LLVM compiler infrastructure and use Clang LibTooling API to instrument target program automatically on AST level.